

A Verified Partial Result for the Vanishing-Merit, Divergent-Gradient Event in Stochastic SQP

Prepared in response to Open Problems in OR entry W3176856692_p0

24 June 2026

Abstract

Berahas, Curtis, O’Neill, and Robinson analyze a stochastic SQP method for equality-constrained optimization with possibly rank-deficient Jacobians. In their vanishing-merit regime, $\tau_k \downarrow 0$, their convergence proof assumes that the realized stochastic-gradient errors $g_k - \nabla f(x_k)$ are eventually uniformly bounded. The complementary event in which $\tau_k \downarrow 0$ while $\sup_k \|g_k - \nabla f(x_k)\| = \infty$ is identified in their paper, and in the Open Problems in OR entry, as an open case.

This note does *not* solve that open problem in full. It gives a corrected and rigorous partial result for the actual Berahas–Curtis–O’Neill–Robinson algorithm. The main point is that the clean pathwise merit decrease assumed in an earlier draft is false for the stochastic algorithm; the correct inequality contains a noise-work term

$$\alpha_k \tau_k (g_k - \nabla f(x_k))^T d_k.$$

If the negative part of this cumulative noise work is summable, and if a small-ball condition ensures enough normally dominated iterations at every persistent infeasibility-stationarity level, then

$$\liminf_{k \rightarrow \infty} \|J_k^T c_k\|_2 = 0,$$

and, for non-diminishing β_k , in fact $\|J_k^T c_k\|_2 \rightarrow 0$. The hypotheses are compatible with unbounded realized errors, for example with Gaussian errors when the vanishing merit weights are summable. The result is therefore genuine partial progress on a nonempty subcase of $E_{\tau, \text{zero}, \text{bad}}$, but it remains conditional and is not a complete resolution of the authors’ open question.

1 Problem, verdict, and notation

Consider

$$\min_{x \in \mathbb{R}^n} f(x) = \mathbb{E}[F(x, \iota)] \quad \text{subject to} \quad c(x) = 0, \tag{1}$$

where $c : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is smooth and $J(x) := \nabla c(x)^T \in \mathbb{R}^{m \times n}$. We use the notation of Berahas–Curtis–O’Neill–Robinson [1]. At iteration k ,

$$c_k = c(x_k), \quad J_k = J(x_k), \quad f_k = f(x_k), \quad e_k = g_k - \nabla f(x_k).$$

The infeasibility-stationarity residual considered here is

$$\theta_k := \|J_k^T c_k\|_2. \tag{2}$$

If $c_k \neq 0$, stationarity of $\varphi(x) = \|c(x)\|_2$ is equivalent to $J_k^T c_k / \|c_k\|_2 = 0$; the residual (2) is the usual stationarity residual for $\frac{1}{2} \|c(x)\|_2^2$ and is one of the residuals explicitly suggested in the open-problem formulation.

The event of interest is

$$E_{\tau, \text{zero}, \text{bad}} := \{\tau_k \downarrow 0\} \cap \left\{ \sup_{k \geq 0} \|g_k - \nabla f(x_k)\|_2 = \infty \right\}. \quad (3)$$

The result below is a *partial* guarantee on subevents of (3). It is not a full solution, because it assumes an additional summability condition on the realized noise work. The value of the result is that it uses the actual one-step estimates of [1], keeps the stochastic error term that the earlier draft omitted, and permits the raw errors $\|e_k\|$ to be unbounded.

2 Algorithmic estimates retained from the original SQP analysis

The SQP step is decomposed as

$$d_k = v_k + u_k, \quad v_k \in \text{Range}(J_k^T), \quad u_k \in \text{Null}(J_k).$$

The merit function and its local model are

$$\begin{aligned} \phi(x, \tau) &= \tau f(x) + \|c(x)\|_2, \\ \ell(x, \tau, g, d) &= \tau(f(x) + g^T d) + \|c(x) + J(x)d\|_2, \end{aligned}$$

with model reduction

$$\Delta \ell_k := \ell(x_k, \tau_k, g_k, 0) - \ell(x_k, \tau_k, g_k, d_k) = -\tau_k g_k^T d_k + \|c_k\|_2 - \|c_k + J_k d_k\|_2. \quad (4)$$

Since $J_k u_k = 0$, $c_k + J_k d_k = c_k + J_k v_k$.

We isolate the deterministic estimates from [1] that are used in the proof.

Assumption 1 (Standing SQP estimates). *The iterates are generated by Algorithm 1 of [1], do not terminate finitely, and satisfy the following estimates.*

- (i) *f is bounded below by $f_{\inf}$ on the iterate-containing set, and there are constants $\kappa_c, \kappa_J, \kappa_g, \kappa_H, \zeta > 0$ such that*

$$\|c_k\|_2 \leq \kappa_c, \quad \|J_k\|_2 \leq \kappa_J, \quad \|\nabla f(x_k)\|_2 \leq \kappa_g, \quad \|H_k\|_2 \leq \kappa_H,$$

and $u^T H_k u \geq \zeta \|u\|_2^2$ for all $u \in \text{Null}(J_k)$.

- (ii) *The normal step satisfies the Cauchy decrease and size estimates: for some $\kappa_v, \omega > 0$,*

$$\|c_k\|_2 (\|c_k\|_2 - \|c_k + J_k v_k\|_2) \geq \kappa_v \|J_k^T c_k\|_2^2, \quad \|v_k\|_2 \leq \omega \|J_k^T c_k\|_2. \quad (5)$$

Consequently,

$$\|c_k\|_2 - \|c_k + J_k v_k\|_2 \geq \kappa_v \kappa_c^{-1} \theta_k^2. \quad (6)$$

- (iii) *The merit-parameter update gives*

$$\Delta \ell_k \geq \tau_k u_k^T H_k u_k + \sigma (\|c_k\|_2 - \|c_k + J_k v_k\|_2) \quad (7)$$

for some $\sigma \in (0, 1)$.

(iv) The step-size safeguard uses parameters $\beta_k \in (0, 1]$ and ξ_k with $\xi_k \geq \xi_{\min} > 0$. If the direction is normally dominated,

$$\|u_k\|_2^2 < \chi_k \|v_k\|_2^2, \quad (8)$$

then

$$\alpha_k \geq a\beta_k, \quad a := \frac{2(1-\eta)\xi_{\min}}{\tau_{\max}L + \Gamma} > 0, \quad (9)$$

where $\tau_{\max}$ is an a priori upper bound on the nonincreasing merit sequence.

Remark 1. The first three items are precisely the normal-step and model-reduction estimates used in [1]. Item (iv) is the normally dominated part of their step-size lower bound. The important correction relative to the earlier draft is that (9) gives a lower bound proportional to β_k , not a uniform positive lower bound unless $\inf_k \beta_k > 0$.

Define the shifted merit quantity

$$S_k := \|c_k\|_2 + \tau_k(f_k - f_{\inf}) \geq 0. \quad (10)$$

The next lemma is the key corrected one-step inequality.

Lemma 1 (Shifted merit inequality with stochastic noise work). *Under Assumption 1, for every iteration k ,*

$$S_k - S_{k+1} \geq \alpha_k(1 - (1 - \eta)\beta_k)\Delta\ell_k + \alpha_k\tau_k e_k^T d_k. \quad (11)$$

In particular, since $\beta_k \in (0, 1]$,

$$S_k - S_{k+1} \geq \eta\alpha_k\Delta\ell_k + \alpha_k\tau_k e_k^T d_k. \quad (12)$$

Proof. The proof is the same descent calculation as Lemma 13 of [1], but without discarding the stochastic-gradient error. That calculation gives

$$\begin{aligned} \|c_k\|_2 - \|c_{k+1}\|_2 &\geq \alpha_k(1 - (1 - \eta)\beta_k)\Delta\ell_k - \tau_k(f_k - f_{k+1}) - \alpha_k\tau_k(\nabla f(x_k) - g_k)^T d_k \\ &= \alpha_k(1 - (1 - \eta)\beta_k)\Delta\ell_k - \tau_k(f_k - f_{k+1}) + \alpha_k\tau_k e_k^T d_k. \end{aligned} \quad (13)$$

Using (10), the monotonicity $\tau_{k+1} \leq \tau_k$, and $f_{k+1} \geq f_{\inf}$,

$$\begin{aligned} S_k - S_{k+1} &= \|c_k\|_2 - \|c_{k+1}\|_2 + \tau_k(f_k - f_{\inf}) - \tau_{k+1}(f_{k+1} - f_{\inf}) \\ &\geq \|c_k\|_2 - \|c_{k+1}\|_2 + \tau_k(f_k - f_{k+1}). \end{aligned}$$

Substituting (13) gives (11). Finally, $1 - (1 - \eta)\beta_k \geq \eta$ for $\beta_k \in (0, 1]$. $\square$

The term

$$W_k := \alpha_k\tau_k e_k^T d_k \quad (14)$$

is the noise work. It is the term absent from the earlier draft. Its negative part is the obstruction to a pathwise monotonicity argument.

3 A pathwise partial result

Let

$$N_k := \mathbf{1}\{\|u_k\|_2^2 < \chi_k \|v_k\|_2^2\}, \quad K_\gamma := \{k : \theta_k \geq \gamma\}.$$

Thus $N_k = 1$ means that the step is normally dominated.

Theorem 1 (Weighted finiteness of normally dominated nonstationary iterations). *Suppose Assumption 1 holds. For every $\gamma > 0$ and every $K \in \mathbb{N}$,*

$$\sum_{\substack{0 \leq k \leq K \\ k \in K_\gamma}} \beta_k N_k \leq \frac{S_0 + \sum_{k=0}^K [W_k]_-}{a\eta\sigma\kappa_v\kappa_c^{-1}\gamma^2}. \quad (15)$$

Consequently, if

$$\sum_{k=0}^{\infty} [\alpha_k \tau_k e_k^T d_k]_- < \infty, \quad (16)$$

then, for every $\gamma > 0$,

$$\sum_{k \in K_\gamma} \beta_k N_k < \infty. \quad (17)$$

Proof. From Lemma 1, (7), and (6),

$$S_k - S_{k+1} \geq \eta\alpha_k\sigma\kappa_v\kappa_c^{-1}\theta_k^2 + W_k. \quad (18)$$

If $k \in K_\gamma$ and $N_k = 1$, then (9) gives

$$S_k - S_{k+1} \geq a\eta\sigma\kappa_v\kappa_c^{-1}\gamma^2\beta_k + W_k. \quad (19)$$

Summing (18) over all $0 \leq k \leq K$, keeping only the selected positive terms and bounding $W_k \geq -[W_k]_-$, yields

$$S_0 - S_{K+1} \geq a\eta\sigma\kappa_v\kappa_c^{-1}\gamma^2 \sum_{\substack{0 \leq k \leq K \\ k \in K_\gamma}} \beta_k N_k - \sum_{k=0}^K [W_k]_-.$$

Since $S_{K+1} \geq 0$, rearranging gives (15). Letting $K \rightarrow \infty$ gives (17) under (16). $\square$

Corollary 1 (Pathwise stationarity consequences). *Assume (16) and $\sum_{k=0}^{\infty} \beta_k = \infty$.*

(a) *If, for every $\gamma > 0$,*

$$\sum_{k \in K_\gamma} \beta_k = \infty \implies \sum_{k \in K_\gamma} \beta_k N_k = \infty, \quad (20)$$

then

$$\liminf_{k \rightarrow \infty} \|J_k^T c_k\|_2 = 0.$$

(b) *If, for every $\gamma > 0$,*

$$|K_\gamma| = \infty \implies \sum_{k \in K_\gamma} \beta_k N_k = \infty, \quad (21)$$

then

$$\|J_k^T c_k\|_2 \rightarrow 0.$$

In particular, (21) follows from $\inf_k \beta_k > 0$ and the statement that every infinite K_γ contains infinitely many normally dominated iterations.

Proof. For (a), if $\liminf_k \theta_k > 0$, then there is $\gamma > 0$ such that K_γ contains all sufficiently large indices. Since $\sum_k \beta_k = \infty$, the left side of (20) holds, so $\sum_{k \in K_\gamma} \beta_k N_k = \infty$, contradicting Theorem 1. For (b), if $\theta_k \not\rightarrow 0$, then K_γ is infinite for some $\gamma > 0$, and (21) again contradicts Theorem 1. $\square$

4 A probabilistic small-ball mechanism

The preceding result is pathwise. We now give a probabilistic condition that implies the normal-domination alternatives in Corollary 1.

Let $\mathcal{F}_k$ be the sigma-field representing the history before the stochastic gradient g_k is sampled, so that $x_k, J_k, c_k, v_k, H_k, \chi_k, \tau_{k-1}$ are $\mathcal{F}_k$ -measurable.

Assumption 2 (Conditional small-ball probability for normal domination). *For every $\gamma > 0$ there exist $p_\gamma > 0$ and $k_\gamma \in \mathbb{N}$ such that, for all $k \geq k_\gamma$,*

$$\mathbb{P}(N_k = 1 \mid \mathcal{F}_k) \geq p_\gamma \quad \text{on } \{\theta_k \geq \gamma\}. \quad (22)$$

Lemma 2 (Weighted conditional Borel–Cantelli). *Let I_k be $\mathcal{F}_k$ -measurable indicators, let $Y_k \in [0, 1]$ be measurable after the k th draw, and let $b_k \in [0, B]$ be deterministic or $\mathcal{F}_k$ -measurable weights. If*

$$\mathbb{E}[Y_k \mid \mathcal{F}_k] \geq p I_k \quad (p > 0),$$

then, almost surely,

$$\sum_{k=0}^{\infty} b_k I_k = \infty \quad \implies \quad \sum_{k=0}^{\infty} b_k I_k Y_k = \infty.$$

Proof. Set $A_n := \sum_{k=0}^n b_k I_k \mathbb{E}[Y_k \mid \mathcal{F}_k]$ and $M_n := \sum_{k=0}^n b_k I_k (Y_k - \mathbb{E}[Y_k \mid \mathcal{F}_k])$. Then M_n is a martingale with increments bounded by B and predictable quadratic variation bounded by BA_n . The martingale strong law for bounded increments gives $M_n/A_n \rightarrow 0$ on $\{A_n \rightarrow \infty\}$. On $\{\sum b_k I_k = \infty\}$, one has $A_n \geq p \sum_{k=0}^n b_k I_k \rightarrow \infty$, and therefore

$$\sum_{k=0}^n b_k I_k Y_k = A_n + M_n \rightarrow \infty.$$

□

Corollary 2 (Almost-sure partial progress on the bad event). *Suppose Assumptions 1 and 2 hold, $\sum_k \beta_k = \infty$, and the error-work summability condition (16) holds almost surely on an event E . Then, almost surely on E ,*

$$\liminf_{k \rightarrow \infty} \|J_k^T c_k\|_2 = 0.$$

If, in addition, $\inf_k \beta_k > 0$, then, almost surely on E ,

$$\|J_k^T c_k\|_2 \rightarrow 0.$$

Consequently, the same conclusions hold on

$$E \cap E_{\tau, \text{zero}, \text{bad}}.$$

Proof. Fix a rational $\gamma > 0$ and apply Lemma 2 with $I_k = \mathbf{1}_{\{\theta_k \geq \gamma\}}$, $Y_k = N_k$, and $b_k = \beta_k$. Assumption 2 implies (20) almost surely. Corollary 1(a) gives the liminf result after intersecting over rational $\gamma > 0$. If $\inf_k \beta_k > 0$, then an infinite set K_γ has $\sum_{k \in K_\gamma} \beta_k = \infty$, so the same argument gives (21); Corollary 1(b) gives convergence. □

5 How the assumptions can hold with unbounded errors

The small-ball condition is natural because the tangential step depends only on the null-space projection of $g_k + H_k v_k$. Let Z_k have orthonormal columns spanning $\text{Null}(J_k)$. The tangential step is

$$u_k = -Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T (g_k + H_k v_k), \quad (23)$$

so

$$\|u_k\|_2 \leq \zeta^{-1} \|Z_k^T (g_k + H_k v_k)\|_2. \quad (24)$$

From (5) and $\|J_k\|_2 \leq \kappa_J$,

$$\theta_k \geq \gamma \implies \|v_k\|_2 \geq \frac{\kappa_v}{\kappa_c \kappa_J} \gamma^2 =: v_\gamma > 0. \quad (25)$$

Hence Assumption 2 is implied by the following lower-tail condition: for each $\gamma > 0$, the conditional law of $Z_k^T (g_k + H_k v_k)$ assigns uniformly positive probability, on $\{\theta_k \geq \gamma\}$, to the ball

$$B(0, \zeta \sqrt{\chi_k} v_\gamma). \quad (26)$$

This condition does not bound the complementary upper tail of g_k .

It remains to explain the noise-work condition. It is not automatic, but it has simple sufficient conditions.

Lemma 3 (A checkable sufficient condition for summable noise work). *Suppose Assumption 1 holds and*

$$\sum_{k=0}^{\infty} \tau_k (1 + \|e_k\|_2^2) < \infty. \quad (27)$$

Then (16) holds.

Proof. The normal-step size bound in Assumption 1 gives $\|v_k\|_2 \leq \omega \|J_k^T c_k\|_2 \leq \omega \kappa_J \kappa_c =: V$. From (23),

$$\begin{aligned} \|u_k\|_2 &\leq \zeta^{-1} \|g_k + H_k v_k\|_2 \\ &\leq \zeta^{-1} (\|e_k\|_2 + \|\nabla f(x_k)\|_2 + \kappa_H V). \end{aligned}$$

Thus there are constants $D_0, D_1 > 0$ such that

$$\|d_k\|_2 \leq D_0 + D_1 \|e_k\|_2 \quad \text{for all } k.$$

Since $0 < \alpha_k \leq 1$,

$$[W_k]_- \leq |W_k| \leq \tau_k \|e_k\|_2 \|d_k\|_2 \leq C \tau_k (1 + \|e_k\|_2^2)$$

for a constant $C > 0$. Summing and using (27) proves the claim. $\square$

Corollary 3 (Gaussian-type compatibility example). *Assume $\inf_k \chi_k > 0$. Assume also that the null-space projected errors have conditional Gaussian laws with covariance eigenvalues uniformly bounded above and below by positive constants, and with conditional means zero. Then the small-ball condition (22) holds. Moreover, the requirements of Corollary 2 are compatible with*

$$\sup_k \|g_k - \nabla f(x_k)\|_2 = \infty.$$

For example, if $\{e_k\}$ are independent nondegenerate Gaussian vectors and $\sum_k \tau_k < \infty$ deterministically, then

$$\sum_k \tau_k (1 + \|e_k\|_2^2) < \infty \quad \text{and} \quad \sup_k \|e_k\|_2 = \infty \quad \text{almost surely.}$$

Proof. The centers $Z_k^T(\nabla f(x_k) + H_k v_k)$ are bounded under Assumption 1. A Gaussian vector with covariance eigenvalues uniformly bounded above and below assigns a uniformly positive probability to every fixed-radius ball centered at any point in a bounded set. This proves the small-ball assertion.

For the final assertion, if $\sum_k \tau_k < \infty$ and the second moments of e_k are uniformly bounded, then

$$\mathbb{E} \left[\sum_k \tau_k (1 + \|e_k\|_2^2) \right] < \infty,$$

so the nonnegative series is finite almost surely. On the other hand, for every $M > 0$ a nondegenerate Gaussian vector satisfies $\mathbb{P}(\|e_k\|_2 > M) > q_M > 0$ uniformly in k ; independence and Borel–Cantelli imply that $\|e_k\|_2 > M$ occurs infinitely often for every integer M , hence $\sup_k \|e_k\|_2 = \infty$ almost surely. $\square$

6 What this fixes, and what it does not fix

The earlier draft contained a useful idea—try to exploit lower-tail events in which the tangential stochastic step is small—but two central assertions were not valid for the actual algorithm of [1].

First, the stochastic algorithm does not provide the clean pathwise decrease

$$S_k - S_{k+1} \geq \eta \alpha_k \Delta \ell_k$$

used in that draft. The correct inequality is Lemma 1, which contains $W_k = \alpha_k \tau_k e_k^T d_k$. Controlling the cumulative negative part of W_k is exactly where unbounded gradient errors matter.

Second, normally dominated steps do not have a stepsize bounded below by a positive constant independent of β_k . The valid lower bound is $\alpha_k \geq a\beta_k$. Therefore the right conclusion is weighted by β_k . With constant β_k , one can get convergence $\theta_k \rightarrow 0$ under the stated hypotheses; with diminishing β_k , the natural conclusion is the liminf guarantee unless additional density of nonstationary normally dominated iterations is assumed.

Thus the present note proves a conditional theorem for the actual algorithm, not for an idealized monotone-merit variant. It gives a rigorous partial answer on

$$E_{\tau, \text{zero}, \text{bad}} \cap \left\{ \sum_k \left[\alpha_k \tau_k e_k^T d_k \right]_- < \infty \right\} \cap \{ \text{the small-ball normal-domination condition holds} \}.$$

It does not prove that the error-work condition follows from the assumptions in [1], nor does it prove that the merit update necessarily makes $\sum_k \tau_k (1 + \|e_k\|^2)$ finite on the bad event. Those are the remaining obstacles to a full solution of the open problem.

7 Conclusion

The corrected argument establishes genuine but conditional partial progress. It proves that, for the Berahas–Curtis–O’Neill–Robinson stochastic SQP algorithm, unbounded realized stochastic-gradient errors do not by themselves preclude infeasibility-stationarity convergence in the vanishing-merit regime. What is needed in the proof is not uniform boundedness of e_k , but control of the accumulated negative noise work $\alpha_k \tau_k e_k^T d_k$, together with enough normally dominated steps. These requirements are compatible with $\sup_k \|e_k\| = \infty$.

The open problem remains unresolved in its full generality: finite second moments and $\tau_k \downarrow 0$ alone do not yet imply the required noise-work summability or the small-ball frequency condition.

References

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